MIL-W-50987 (MU)
14 June 1972
Superseding
PA-PD-2581 (Rev. 1)
23 August 1965

SPECIFICATION

WARHEAD, 2.75 INCH ROCKET, HE, M151 WITH FUZE OR WITHOUT FUZE, LOADING, ASSEMBLING PACKING

1. SCOPE

1.1 This specification covers the quality assurance Provisions and special requirements not covered by the drawings; for one type of high explosive warhead designated as Warhead, 2.75 Inch Rockets HE, M151.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons, General Specification For.

STANDARDS

MILITARY

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109 - Quality Assurance Terms & Definitions.
MIL-STD-1168- Lot Numbering of Ammunition.
MIL-STD-1235- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

FSC: 1340

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DRAWINGS

ORDNANCE CORPS

8882186	Warhead, 2.75 Inch Rocket, HE, M151 w/Fuze or W/O Fuze.
8886810	Box, Packing, Ammunition, for Warhead, 2.75 Inch Rocket, HE, M151, Smoke WP M156 and Practice XM230 and XM232.
8886811	Container, Ammo, Fiber, M523 for Warhead, 2.75 Inch Rocket, HE, M151, Smoke WP M156 and Practice XM230 & XM232.

PUBLICATIONS

ORDNANCE CORPS

ET-8882186 - Equipment Tabulation.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

3. Requirements.

3.1 Material. - Materials shall be in accordance with applicable drawings and specifications.

3.2 Loaded Assemblies. - The loaded assemblies shall comply with all of the requirements specified on Drawing (Dwg.) 8882186 and with all requirements specified in applicable specifications.

3.3 Cavitation in Comp B-4.

3.3.1 There shall be no evidence of cavitation in the faced Comp B-4 greater than the cavitation allowed in 3.3.2, 3.3.3, 3.3.4, and 3.3.5.

3.3.2 Cracked Charge. - A charge shall be considered cracked when a fissure one thirtysecond of an inch (1/32 in.) or less in width appears on the faced surface with no visual appearance of widening beneath the surface. An acceptable warhead shall contain no more than two (2) such cracks.

3.3.3 Piping Cavities. - Any cavity having a dimension greater than one eighth inch (1/8 in.) at the faced surface and extending to a depth of greater than one half inch (1/2 in.) below

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the faced surface shall be termed a pipe. Acceptable warheads shall contain no pipes having a dimension greater than one quarter in.(1/4 in.) on the faced surface. Warheads containing pipes which are larger beneath the faced surface than at the surface shall be unacceptable.

3.3.4 Surface Cavities. - Surface cavities are those cavities having a dimension greater than one eighth inch (1/8 in.) at the faced surface which are neither cracks nor pipes as defined in 3.3.2 and 3.3.3. There shall be no surface cavities having a dimension greater than one quarter (1/4 inch.) on the faced surface.

3.3.5 Blemishes. - Any cavity not having a dimension at the faced surface greater than one eighth inch (1/8 in.) shall be considered a blemish. Blemishes shall not be used to determine acceptability of a warhead.

3.4 Workmanship. - Loading and assembling shall be accomplished in accordance with best practices. All parts shall be free of any foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Surface coatings shall be continuous except for a few slight scratches not exposing base material. All required marking shall be neat and sharply defined.

3.5 First Article Inspection. - This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspection are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to defined terms used herein. The provisions of MIL-A-2550 shall apply.

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4.1,1 Submission of Product. - At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with Identification and date of changes, thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material Is controlled by Government or commercial specifications referenced In any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent’s authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First Article Sample.

4.2.1 Submission, - The contractor shall submit a first article sample to a Government approved facility as designated by the Contracting Officer (see para, 6.4) for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of 10 assemblies and 10 complete sets of parts (i.e., 10 each of every component, and every subassembly) which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production, Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, Including statements of findings, certificates of conformance for materials (see 4.1.1). A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer. A first article sample, or portion thereof, as directed by the Contracting Officer, shall also be submitted

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whenever there is a lapse in production for a period in excess of 9(3 days or whenever a change occurs in manufacturing, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inspections to be Performed. - Assemblies and components will be subjected by the Government to any or all of the examinations or tests specified in this specification and any or all requirements of the drawings.

4.2.3 Rejection. - If any assembly or component fails to comply with any of the requirements and inspections, the first article sample shall be rejected. The (government reserves the right to terminate its inspection upon any failure of an assembly or component in the sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and/or submit a new first article sample or portion thereof. Until a first article sample is accepted the contractor is, in no way, authorized by the Government to initiate regular production unless directed by the Contracting Officer.

4.3 Inspection Provisions for Production.

4.3.1 Lot Formation. - The term “inspection lot” as used throughout this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one man== ufacturer, in one unchanged process, in accordance with the same drawings same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105, Changes to the process, specification or drawing not affecting safety, performances interchangeability, or storages as determined by the Government, shall not be deemed to alter the homogeneity of the inspection lot. All material submitted for Inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot shall bear a unique lot number. Each lot shall consist of:

a. Metal parts from lots bearing the same lot Interfix number

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from one manufacturer.
b. Composition B-4 from not more than three lots with the same interfix number from one manufacturer.

4.3.2 Examination. - The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies in the applicable paragraph headings. Inspection for critical defects (and major defects when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 nay be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply the AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspection listed shall be in accordance with 4.3.4.

4.3.2.1 Metal Parts Assembly (see dwg. 8889406 covering a detail of dwg. 8882186).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Interior coating damaged or not continuous	Visual	01001
102.	Foreign matter present in interior	Visual	01002
Minor:	None defined.		

4.3.2.2 Warhead, loaded, prior to insertion of fuze cavity liner or assembly of fuze (see dwg. 8882186).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		

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Major:	AQL 0.65 percent		
101.	Charge missing	Visual	02001
102.	Explosive in thread	Visual	02002
103.	Distance to charge	Gage	02003
104.	Sealing compound missing (if to be fuzed)	Visual	20004
Minor:	AQL 0.65 percent		
201.	Silicone compound missing from base thread	Visual	02005

4.3.2.3 Warhead, loaded, after insertion of fuze cavity liner or assembly of fuze.

4.3.2.3.1 Warhead, loaded, after insertion of fuze cavity liner, if to be shipped without fuze assembled (see dwg.8882186).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Fuze cavity liner missing or improperly inserted	Visual	03001
102.	Distance to fuze cavity liner	Gage	03002
103.	Anti-sieze compound missing from fuze thread or incomplete coverage	Visual	03003
104.	Thread damaged	Visual	03004
105.	Bourrelet damaged	Visual	03005
106.	Diameter of bourrelet, max.	Gage	03006
Minor:	AQL 1.00 percent		
201.	Marking missing, misleading or unidentifiable	Visual	03007
202.	Evidence of poor workmanship (see 3.4)	Visual	03008

4.3.2.3.2 Warhead, loaded (see dwg. 8882186).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Fuze type improper	Visual	03009

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Major:	AQL 0.65 percent		
101.	Thread damaged	Visual	03004
102.	Bourrelet damaged	Visual	03005
103.	Diameter of bourrelet, max	Gage	03006
Minor:	AQL 1.00 percent		
201.	Marking missing, misleading or unidentifiable	Visual	03007
202.	Evidence of poor workmanship (see 3.4)	Visual	03008

4.3.2.4 Fiber container, sealed, prior to taping (see dwg. 8886811).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major: None defined.			
Minor:	AQL 0.65 percent		
201.	Gap between body and cover in excess of 1/8 inch	Gage	04001

4.3.2.5 Fiber container, sealed (see dwg. 8886811).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.65 percent		
101.	Tape incomplete or badly wrinkled	Visual	05001
102.	Cuts, scuffs, gouges or checks which penetrate outer layer	Visual	05002
103.	Metal end loose or distorted	Visual/Manual	05003
Minor:	AQL 1.50 percent		
202.	Marking missing, misleading or unidentifiable	Visual	05005
203.	Contents loose	Manual	05006

4.3.2.6 Box, shipping (see dwg. 8886810).

Categories	Defects	Method of Inspection	Code No.

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Critical: None defined.

Major:	AQL 1.00 percent		
101.	Contents exposed	Visual	06001
102.	Board broken	Visual	
103.	FSC number missing or misleading	Visual	06003
104.	Hardware missing, broken or loose	Visual/Manual	06004
105.	Strapping missing, broken or loose	Visual/Manual	06005
Minor:	AQL 1.50 percent		
201.	Marking, misleading or unidentifiable	Visual	06006
202.	Hardware improperly engaged	Visual	06007
203.	Car seal missing, unsealed or improperly positioned	Visual	06008
204.	Handle missing or insecure	Visual/Manual	06009

4.3.3 Testing.

4.3.3.1 Cavitation (see 3.3) - Major defect - Code No. 02005. Each lot of warheads, loaded, prior to insertion of the fuze cavity liner or assembly of the fuze shall be one hundred percent (100%) subjected to the sequence specified in 4,4.1 using equipment in accordance with 4.3.4. Units with more than two cracks (3.3.2) shall be considered defective. Units with pipes (3.3.3) which appear to widen beneath the surface shall be considered defective. Units with any oversize surface cavity (3.3.4) shall be considered defective.

4.3.4 Inspection Equipment. - Equipment Tabulation ET-8882186 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.3.4.1 Government Rights to Documentation. - Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, constructions survey and wherever inspection equipment drawings are needed.

4.4 Test Methods.

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4.4,1 Determination of Cavities. - The warhead, loaded, prior to insertion of the fuze cavity liner or assembly of the fuze shall be visually inspected using a comparison scale. Warheads with no more than two cracks (3.3.2) and any cavity greater than a blemish (3.3.5) but smaller than an oversize surface cavity (3.304) shall be probed with a one sixteenth inch (1/16 in.) minimum diameter probe of suitable material. If probing indicates a pipe (3.3.3) the cavity shall be visually examined for widening of the pipe below the surface.

5. PREPARATION FOR DELIVERY.

5.1 Packaging.

5.1.1 Level A. - The warheads, with or without fuzes, shall be packaged in accordance with dwg. 8886811,

5.2 Packing.

5.2.1 Level A. - The packaged assemblies shall be packed and marked in accordance with dwg. 8886810.

6. NOTES

6.1 Ordering Data. - Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Ammunition data cards shall be prepared for each lot in accordance with MIL-STD-1167.

6.2 Inspection Code Numbers. - The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government,

6.3 Inspection Equipment. - The contractor shall design inspection equipment as required by the referenced inspection Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection Equipment Lists. - Inspection equipment lists indicate the availability of inspection equipment designs by showing in the “number” column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3,4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for

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all the equipment coded as “Contractor Design” in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection Equipment List codes. - The inspection equipment as defined in 6.3.3 and 6.3.11 will be designated in the EL by the Following codes:

CD - Contractor design 01) controlled contractor format arid/or commercial equipment.
MU - Amy design, mandatory for use.

6.3.3 Army Designs. - Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated “MU”) and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of’ the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor Designs. - Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions . Contractors shall submit one copy of final designs as a rolled set. Designs shall be submitted for approval as specified in 6.3.5.

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6,3.5 Submission of Designs. - Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-QA-A-R. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the Contracting Officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.4 Submission of First Article Samples. - Instruction as to the location for the evaluation of the first article sample shall be obtained from the Contracting Officer. Upon receipt of such request, the Contracting Officer shall advise Picatinny Arsenal and instructions will be issued accordingly. All inquiries should be forwarded to: Commanding officer, Picatinny Arsenal, ATTN: SMUPA-QA-I,

6.5 The warhead-fuze combination is ballistically tested in accordance with specification MIL-F-50253 (MU), Fuzes, Rocket, M423 and M427 Loading, Assembling and Packing.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodian:
Army - NJ

Preparing Activity:
Army - MU

Project No.: 1340

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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Commanding Officer
Picatinny Arsenal
ATTN: SMUPA-[redacted] G·A-A
Dover, New Jersey 07801

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